

REPORT FILE

```
#include<stdio.h>
#include<string.h>

struct student {
    long int id;
    char nm[50];
    char mail[50];
    char ph[10];
    int yr;
    int sm;
    float sg[8];
    float cg;
};

struct student s[5];
int n = 0;

//function to calculate cgpa
float calc(float sg[], int sm) {
    float sum = 0;
    int x = sm - 1;

    if (x <= 0) return 0;

    for (int i = 0; i < x; i++)
        sum += sg[i];

    return sum / x;
}

//add student
void add() {
    if (n == 5) {
        printf("\nLIMIT FULL!\n");
        return;
    }

    printf("SAP ID: ");
    scanf("%ld", &s[n].id);

    printf("Name: ");
    scanf(" %[^\\n]s", s[n].nm);

    printf("Email: ");
    scanf(" %[^\\n]s", s[n].mail);

    printf("Phone: ");
```

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    scanf("%[^\\n]s", s[n].ph);
printf("Year: ");
    scanf("%d", &s[n].yr);

printf("Semester: ");
scanf("%d", &s[n].sm);

for (int i = 0; i < 8; i++) {
    if (i < s[n].sm - 1) {
        printf("SGPA sem %d: ", i + 1);
        scanf("%f", &s[n].sg[i]);
    } else {
        s[n].sg[i] = 0;
    }
}

s[n].cg = calc(s[n].sg, s[n].sm);

n++;
printf("\\nStudent added!\\n");
}

```

```

//search
void find() {
    long int idd;
    int ok = 0, p = -1;

printf("Enter SAP ID: ");
scanf("%ld", &idd);

for (int i = 0; i < n; i++) {
    if (idd == s[i].id) {
        ok = 1;
        p = i;
        break;
    }
}

if (ok) {
printf("\\n-----\\n");
printf("SAP ID: %ld\\n", s[p].id);
printf("Name: %s\\n", s[p].nm);
printf("Phone: %s\\n", s[p].ph);
printf("Email: %s\\n", s[p].mail);
printf("Year: %d\\n", s[p].yr);
printf("Semester: %d\\n", s[p].sm);

for (int j = 0; j < 8; j++)
    printf("SGPA SEM %d = %.2f\\n", j + 1, s[p].sg[j]);
}
}

```

```

        printf("CGPA: %.2f\n", s[p].cg);
    } else {
        printf("\nNo record found!\n");
    }
}

```

```

//avg CGPA
void avg() {
    int y, c = 0;
    float sum = 0;

    printf("Enter year: ");
    scanf("%d", &y);

    for (int i = 0; i < n; i++) {
        if (y == s[i].yr) {
            sum += s[i].cg;
            c++;
        }
    }

    if (c == 0)
        printf("\nNo record in year %d\n", y);
    else
        printf("AVG CGPA of year %d = %.2f\n", y, sum / c);
}

```

```

//show all
void show() {

    if (n == 0) {
        printf("\nNO RECORDS!\n");
        return;
    }

    for (int i = 0; i < n; i++) {
        printf("\n-----\n");
        printf("SAP ID: %ld\n", s[i].id);
        printf("Name: %s\n", s[i].nm);
        printf("Phone: %s\n", s[i].ph);
        printf("Email: %s\n", s[i].mail);
        printf("Year: %d\n", s[i].yr);
        printf("Semester: %d\n", s[i].sm);

        for (int j = 0; j < 8; j++)
            printf("SGPA SEM %d = %.2f\n", j + 1, s[i].sg[j]);

        printf("CGPA: %.2f\n", s[i].cg);
    }
}

```

```
    }  
}
```

```
//delete student  
void del() {  
    long int idd;  
    int ok = 0;  
  
    printf("Enter SAP ID: ");  
    scanf("%ld", &idd);  
  
    for (int i = 0; i < n; i++) {  
        if (idd == s[i].id) {  
            ok = 1;  
            for (int j = i; j < n - 1; j++)  
                s[j] = s[j + 1];  
            n--;  
            printf("Record deleted!\n");  
            break;  
        }  
    }  
  
    if (!ok)  
        printf("No record found\n");  
}
```

```
//sort by SAP ID  
void sortid() {  
    struct student t;  
  
    for (int i = 0; i < n - 1; i++) {  
        for (int j = i + 1; j < n; j++) {  
            if (s[j].id < s[i].id) {  
                t = s[i];  
                s[i] = s[j];  
                s[j] = t;  
            }  
        }  
    }  
  
    printf("\nSorted by SAP ID!\n");  
    show();  
}
```

```
//sort by name  
void sortnm() {
```

```

    struct student t;

    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (strcmp(s[j].nm, s[i].nm) < 0) {
                t = s[i];
                s[i] = s[j];
                s[j] = t;
            }
        }
    }

    printf("\nSorted by Name!\n");
    show();
}

int main() {
    int ch;

    while (1) {
        printf("\n----- MENU ----- \n");
        printf("1. Add Student\n");
        printf("2. Search Student\n");
        printf("3. Display All\n");
        printf("4. Delete Student\n");
        printf("5. Avg CGPA of Year\n");
        printf("6. Sort by SAP ID\n");
        printf("7. Sort by Name\n");
        printf("8. Exit\n");
        printf("----- \n");

        printf("Enter Choice: ");
        scanf("%d", &ch);

        switch (ch) {
            case 1: add(); break;
            case 2: find(); break;
            case 3: show(); break;
            case 4: del(); break;
            case 5: avg(); break;
            case 6: sortid(); break;
            case 7: sortnm(); break;
            case 8: return 0;
            default: printf("Invalid choice!\n");
        }
    }
}

```

OUTPUT:

```
4. Delete Student
5. Avg CGPA of Year
6. Sort by SAP ID
7. Sort by Name
8. Exit
```

```
-----
Enter Choice: 1
SAP ID: 590028303
Name: shreya
Email: shreya@gmail.com
Phone: 9958771912
Year: 3
Semester: 5
SGPA sem 1: 8
SGPA sem 2: 8
SGPA sem 3: 8
SGPA sem 4: 8
```

Student added!

```
----- MENU -----
1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Avg CGPA of Year
6. Sort by SAP ID
7. Sort by Name
8. Exit
```

```
-----
Enter Choice: 1
SAP ID: 590027352
Name: anahita
Email: anahuta@gmail.com
Phone: 9045037739
Year: 2
Semester: 3
SGPA sem 1: 8
SGPA sem 2: 9
```

Student added!

----- MENU -----

1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Avg CGPA of Year
6. Sort by SAP ID
7. Sort by Name
8. Exit

Enter Choice: 1

SAP ID: 590027682

Name: vansh

Email: vansh@fml

Phone: 7982299076

Year: 3

Semester: 5

SGPA sem 1: 8

SGPA sem 2: 8

SGPA sem 3: 8

SGPA sem 4: 9

Student added!

----- MENU -----

1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Avg CGPA of Year
6. Sort by SAP ID
7. Sort by Name
8. Exit

Enter Choice: 2

Enter SAP ID: 590028303

SAP ID: 590028303

Name: shreya

Phone: 9958771912


```
-----  
SAP ID: 590028303  
Name: shreya  
Phone: 9958771912  
Email: shreya@gmail.com  
Year: 3  
Semester: 5  
SGPA SEM 1 = 8.00  
SGPA SEM 2 = 8.00  
SGPA SEM 3 = 8.00  
SGPA SEM 4 = 8.00  
SGPA SEM 5 = 0.00  
SGPA SEM 6 = 0.00  
SGPA SEM 7 = 0.00  
SGPA SEM 8 = 0.00  
CGPA: 8.00
```

```
----- MENU -----  
1. Add Student  
2. Search Student  
3. Display All  
4. Delete Student  
5. Avg CGPA of Year  
6. Sort by SAP ID  
7. Sort by Name  
8. Exit
```

```
-----  
Enter Choice: 3
```

```
-----  
SAP ID: 590028303  
Name: shreya  
Phone: 9958771912  
Email: shreya@gmail.com  
Year: 3  
Semester: 5  
SGPA SEM 1 = 8.00  
SGPA SEM 2 = 8.00  
SGPA SEM 3 = 8.00  
SGPA SEM 4 = 8.00  
SGPA SEM 5 = 0.00
```

Year: 3
Semester: 5
SGPA SEM 1 = 8.00
SGPA SEM 2 = 8.00
SGPA SEM 3 = 8.00
SGPA SEM 4 = 8.00
SGPA SEM 5 = 0.00
SGPA SEM 6 = 0.00
SGPA SEM 7 = 0.00
SGPA SEM 8 = 0.00
CGPA: 8.00

SAP ID: 590027352
Name: anahita
Phone: 9045037739
Email: anahuta@gmail.com
Year: 2

Semester: 3
SGPA SEM 1 = 8.00
SGPA SEM 2 = 9.00
SGPA SEM 3 = 0.00
SGPA SEM 4 = 0.00
SGPA SEM 5 = 0.00
SGPA SEM 6 = 0.00
SGPA SEM 7 = 0.00
SGPA SEM 8 = 0.00
CGPA: 8.50

SAP ID: 590027682
Name: vansh
Phone: 7982299076
Email: vansh@fmil
Year: 3

Semester: 5
SGPA SEM 1 = 8.00
SGPA SEM 2 = 8.00
SGPA SEM 3 = 8.00
SGPA SEM 4 = 9.00
SGPA SEM 5 = 0.00

```
-----  
SAP ID: 590027682  
Name: vansh  
Phone: 7982299076  
Email: vansh@fmil  
Year: 3  
Semester: 5  
SGPA SEM 1 = 8.00  
SGPA SEM 2 = 8.00  
SGPA SEM 3 = 8.00  
SGPA SEM 4 = 9.00  
SGPA SEM 5 = 0.00  
SGPA SEM 6 = 0.00  
SGPA SEM 7 = 0.00  
SGPA SEM 8 = 0.00  
CGPA: 8.25
```

```
----- MENU -----  
1. Add Student  
2. Search Student  
3. Display All  
4. Delete Student  
5. Avg CGPA of Year  
6. Sort by SAP ID  
7. Sort by Name  
8. Exit
```

```
-----  
Enter Choice: 4  
Enter SAP ID: 590027352  
Record deleted!
```

```
----- MENU -----  
1. Add Student  
2. Search Student  
3. Display All  
4. Delete Student  
5. Avg CGPA of Year  
6. Sort by SAP ID  
7. Sort by Name  
8. Exit
```

```
-----  
Enter Choice: 2  
Enter SAP ID: 590027352
```

No record found!

```
----- MENU -----  
1. Add Student  
2. Search Student  
3. Display All  
4. Delete Student  
5. Avg CGPA of Year  
6. Sort by SAP ID  
7. Sort by Name  
8. Exit  
-----
```

```
Enter Choice: 5  
Enter year: 3  
AVG CGPA of year 3 = 8.12
```

```
----- MENU -----  
1. Add Student  
2. Search Student  
3. Display All  
4. Delete Student  
5. Avg CGPA of Year  
6. Sort by SAP ID  
7. Sort by Name  
8. Exit  
-----
```

```
Enter Choice: 6  
  
Sorted by SAP ID!
```

```
-----  
SAP ID: 590027682  
Name: vansh  
Phone: 7982299076  
Email: vansh@fmil  
Year: 3  
Semester: 5
```

```
-----  
SAP ID: 590027682  
Name: vansh  
Phone: 7982299076  
Email: vansh@fml  
Year: 3  
Semester: 5  
SGPA SEM 1 = 8.00  
SGPA SEM 2 = 8.00  
SGPA SEM 3 = 8.00  
SGPA SEM 4 = 9.00  
SGPA SEM 5 = 0.00  
SGPA SEM 6 = 0.00  
SGPA SEM 7 = 0.00  
SGPA SEM 8 = 0.00  
CGPA: 8.25
```

```
-----  
SAP ID: 590028303  
Name: shreya  
Phone: 9958771912  
Email: shreya@gmail.com  
Year: 3  
Semester: 5  
SGPA SEM 1 = 8.00  
SGPA SEM 2 = 8.00  
SGPA SEM 3 = 8.00  
SGPA SEM 4 = 8.00  
SGPA SEM 5 = 0.00  
SGPA SEM 6 = 0.00  
SGPA SEM 7 = 0.00  
SGPA SEM 8 = 0.00  
CGPA: 8.00
```

```
----- MENU -----  
1. Add Student  
2. Search Student  
3. Display All  
4. Delete Student  
5. Avg CGPA of Year  
6. Sort by SAP ID
```

----- MENU -----

1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Avg CGPA of Year
6. Sort by SAP ID
7. Sort by Name
8. Exit

Enter Choice: 7

Sorted by Name!

SAP ID: 590028303
Name: shreya
Phone: 9958771912
Email: shreya@gmail.com
Year: 3
Semester: 5
SGPA SEM 1 = 8.00
SGPA SEM 2 = 8.00
SGPA SEM 3 = 8.00
SGPA SEM 4 = 8.00
SGPA SEM 5 = 0.00
SGPA SEM 6 = 0.00
SGPA SEM 7 = 0.00
SGPA SEM 8 = 0.00
CGPA: 8.00

SAP ID: 590027682
Name: vansh
Phone: 7982299076
Email: vansh@fmil
Year: 3
Semester: 5
SGPA SEM 1 = 8.00
SGPA SEM 2 = 8.00

```
SGPA SEM 2 = 8.00
SGPA SEM 3 = 8.00
SGPA SEM 4 = 8.00
SGPA SEM 5 = 0.00
SGPA SEM 6 = 0.00
SGPA SEM 7 = 0.00
SGPA SEM 8 = 0.00
CGPA: 8.00
```

```
-----
SAP ID: 590027682
Name: vansh
Phone: 7982299076
Email: vansh@fmil
Year: 3
Semester: 5
SGPA SEM 1 = 8.00
SGPA SEM 2 = 8.00
SGPA SEM 3 = 8.00
SGPA SEM 4 = 9.00
SGPA SEM 5 = 0.00
SGPA SEM 6 = 0.00
SGPA SEM 7 = 0.00
SGPA SEM 8 = 0.00
CGPA: 8.25
```

```
----- MENU -----
1. Add Student
2. Search Student
3. Display All
4. Delete Student
5. Avg CGPA of Year
6. Sort by SAP ID
7. Sort by Name
8. Exit
```

```
-----
Enter Choice: 8
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```